

Mohamed Waleed

Software Engineer — Flutter Developer
Cairo, Egypt

Mohamedwaleed200155@gmail.com

+20 1131243723

github.com/mohamed-waleed55-tech

linkedin.com/in/mohamed-waleed-87aa151bb

PROFESSIONAL SUMMARY

Software Engineer specializing in Flutter with experience designing scalable, maintainable, and high-performance mobile applications. Strong foundation in software engineering principles including Clean Architecture, SOLID, Design Patterns, data structures, algorithms, and object-oriented programming. Experienced in building offline-first applications, integrating REST APIs and Firebase services, and applying modern software development practices to deliver reliable production-ready solutions. Passionate about solving complex engineering problems and continuously improving software quality.

TECHNICAL SKILLS

- **Languages:** Dart, Kotlin, Java, C++, SQL, Python
- **Mobile Development:** Flutter, Native Android (Kotlin), Clean Architecture, MVVM, MVC, Responsive UI, Unit Testing
- **State Management:** GetX, BLoC/Cubit, Hydrated BLoC, Riverpod, Provider, InheritedWidget, Dependency Injection (Get_it, Injectable)
- **Backend & APIs:** RESTful APIs, WebSockets, Firebase (Authentication, Firestore, Cloud Storage, Crashlytics, Analytics), Google Gemini API, AI Integration, JSON Parsing, Dio, Retrofit, Freezed, API Error Handling
- **Databases & Storage:** SQFlite, Hive, SharedPreferences, GetStorage, Offline Synchronization, Local Caching
- **Software Engineering:** OOP, SOLID Principles, Design Patterns, Repository Pattern, Data Structures, Algorithms, CI/CD Pipelines, Flavors, Shorebird
- **AI & Developer Productivity:** Prompt Engineering, AI-assisted Development, AI Agents, LLM Integration, AI-powered Search, Code Generation & Refactoring
- **Tools & Methodologies:** Git, GitHub, Android Studio, VS Code, Postman, Figma, Agile/Scrum, Crashlytics, Sentry

FEATURED PROJECTS

- **EduLink – Student Social Platform** Graduation Project (Grade: 196/200)
Tech: Kotlin Multiplatform, Ktor, Supabase, Firebase, Room 2024
 - Engineered a cross-platform academic social ecosystem using Kotlin Multiplatform to encapsulate core business logic across target environments.
 - Architected real-time communication modules including one-to-one chats, global communities, multi-group network feeds, and live collaborative academic tools.
 - Integrated AI-driven computer vision and text processing models to enforce community guidelines by filtering and detecting inappropriate content dynamically.
 - Leveraged Kotlin Coroutines for safe asynchronous operations and Room for local database synchronization under a strict MVVM layout.
- **ClickUP E-Commerce Platform App** GitHub Repository
Tech: Flutter, Gemini API, DummyJSON API, GetX, Retrofit, Dio, Freezed, SQFlite, Firebase, Clean Architecture 2026
 - Developed an offline-first E-Commerce application using Flutter, MVVM, and GetX for scalable state management.
 - Integrated Gemini API to create an AI shopping assistant that understands user intent, extracts product requirements, and improves product discovery.
 - Implemented a repository-based data layer integrating DummyJSON API, Firebase, and SQFlite for online/offline product browsing and cart synchronization.
 - Built complete shopping features including authentication, categories, best sellers, favorites, cart, checkout, and user profile management.
 - Applied secure networking with Dio/Retrofit, Freezed models, Crashlytics monitoring, API error handling, and environment-based configuration.
- **Eventra – Real-Time Event Management Platform** GitHub Repository
Tech: Flutter, Firebase, BLoC/Cubit, Google Maps API, MVVM Architecture 2025

- Developed a real-time event management application using Flutter, MVVM, and BLoC/Cubit for scalable state management and clean separation of concerns.
- Integrated Google Maps API to provide location-based event discovery with interactive maps and geolocation features.
- Implemented Firebase Authentication and Firestore for secure user management and real-time event synchronization.
- Built responsive UI features including localization, dynamic themes, and optimized local data persistence for improved user experience.

• Global News Platform

GitHub Repository

Tech: Flutter, REST API, BLoC/Cubit, MVVM Architecture

2026

- Built a real-time news aggregation engine making concurrent REST API polling calls equipped with intuitive category filtration.
- Decoupled UI components from network models using an MVVM architecture pattern to ensure robust testability and responsive state transitions.

PROFESSIONAL TRAINING

Information Technology Institute (ITI)

Egypt

Kotlin Developer Intern

2023

- * **Native Android Systems:** Authored Android application services using native Kotlin development structures under a strict MVVM layout alongside Room relational databases.
- * **Concurrency Control:** Implemented asynchronous tasks safely using Kotlin Coroutines to maintain 60FPS thread rendering operations.

Information Technology Institute (ITI)

Egypt

Java Developer Intern

2022

- * **Core Architecture:** Solidified OOP theory, multithreading control setups, and core system architectures inside clean standalone Java environments.
- * **SOLID Compliance:** Refactored legacy code blocks to strictly comply with SOLID principles to minimize technical debt and elevate maintainability metrics.

EDUCATION

Helwan University

Cairo, Egypt

Bachelor of Engineering in Computer and Systems Engineering; Grade: Good

2019 – 2024

- * **Key Coursework:** Advanced Data Structures, Analysis of Algorithms, OOP, Database Management Systems, Operating Systems Architecture.

PROFESSIONAL DEVELOPMENT & READING

Clean Code – Robert C. Martin

A Handbook of Agile Software Craftsmanship

- * Studied and applied principles of writing readable, maintainable, and refactorable code, focusing on meaningful naming, solid functions, and proper error handling.

Head First Design Patterns – O'Reilly

Building Extensible and Maintainable Object-Oriented Software

- * Mastered core Object-Oriented design patterns (such as Strategy, Observer, Factory, and Singleton) to build flexible, decoupled software architectures.

Grokking Algorithms – Aditya Bhargava

An Illustrated Guide for Programmers and Other Curious People

- * Built a strong foundation in foundational computer science concepts, including algorithm runtime analysis (Big O), sorting, binary search, and trees.